

Title: Design & Implementation of a Multi-Campus University Network Infrastructure

Semester Project Outline

Course: Computer Networks

Semester: BSSE-6th

Project Timeframe: 4 weeks

Team Size: 3 students

Project Objective

Students will design and simulate the complete network infrastructure of a university having six (06) geographically distributed campuses across the country. They must implement IP addressing, subnetting, routing protocols, WAN connectivity, security layers, and redundancy mechanisms using a network simulation tool such as Cisco Packet Tracer.

Project Requirements

1. Campus Overview

Each student/team must assume the university has 6 campuses located in:

- Campus A (Head Office / Data Center)
- Campus B (Northern Region)
- Campus C (Southern Region)
- Campus D (Eastern Region)
- Campus E (Western Region)
- Campus F (Central Region)

All the Campuses should have a different numbers of Students and Faculty member like in the given below table.

Department	PCs	Servers (DHCP, DNS, Web, Mail)	Network Devices
Admin	20-40	1	1 or more Switches
Faculty	50 to 100	1	1 or more Switches
Students	500 to 1000	1	1 or more Switches
Wi-Fi Zone	10 - 50 APs	-	Wireless LAN Controller (optional)

2. IP Addressing & Subnetting

- Use a given public IP block (e.g., 10.0.0.0/8 or 200.100.0.0/16)
- Perform VLSM / FLISM-based subnetting
- Allocate separate VLANs for Admin, Faculty, Student, and Server networks

3. Inter-Campus Connectivity

- Connect campuses using either MPLS, VPN, or Serial WAN Links
- Define Primary & Backup WAN Links for redundancy
- Create a logical topology diagram (Hierarchical / Mesh / Hybrid)

4. Routing Implementation

Students must configure at least one of the following combinations:

Scenario	Interior Protocol	Exterior Protocol
1	RIP / OSPF	Static
2	OSPF (Single or Multi-Area)	EIGRP
3	EIGRP	BGP (for ISP Simulation)

5. Network Services

- DHCP for automatic IP allocation
- DNS & Web Server (simulated)
- Access Control Lists (ACLs) to restrict access between VLANs
- NAT for internet access simulation
- **Optional:** Load balancing / Firewall Integration

6. Network Security & Redundancy (Optional Bonus)

- Configure Port Security / WPA2 Enterprise for Wi-Fi
- Apply HSRP / VRRP for router failover
- Implement Switch STP / RSTP to prevent loops

Project Deliverables

- 1. Network Topology Diagram - Physical + Logical
- 2. IP Addressing & Subnetting Table - Including VLSM plans
- 3. Routing & VLAN Configuration Files - All CLI commands
- 4. Simulation File - .pkt / .gns3 / .eve
- 5. Final Report - PDF with documentation and screenshots
- 6. Live Demonstration - Simulated pings and routing tables

Evaluation Criteria	
Component	Marks
Topology Design & Planning	20
Correct IP Addressing & Subnetting	15
Routing Implementation	25
Network Services & Security	20

Documentation & Presentation	20